Data-Driven Insights from a Real-World Public Report

Title: **Case Study: Crime Against Women in India (2018–2022) – A Data-Driven Analysis Using Power BI**

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Abstract:

The present analytics project aims to analyse the trends, regional patterns, and extent of crimes against women in India, relying entirely on the freely accessible government data. The source of the data is the District-wise Crime Against Women Report (2017 onwards), containing the main indicators such as rape, kidnapping/abduction, dowry deaths, domestic violence, and total reported cases for all Indian districts and states. The cleaned, modelled, and transformed raw CSV file was turned into an interactive dashboard that includes KPIs, trend lines, geographical maps, and ranking charts using Power BI. One of the main goals of the research is to assess the growth of crime trends over the years, make comparisons among states and districts, locate hotspots, and point out the areas with urgent policy requirements. The analysis points out a significant difference in the situation of different regions, as some states have been recording an increase in crime rates while others have succeeded in reducing theirs through specific interventions. This project indicates the worth of public reports when converted into actionable insights through business intelligence tools, hence assisting policymakers, law enforcement agencies, and social institutions in the process of safety monitoring and corrective measures implementation.

1. Introduction:

In India, the persecution of women is still very much alive and is considered to be a major social and governance problem. The National Crime Records Bureau (NCRB) is the government agency that provides annual statistics reflecting the occurrence of such crimes as rape, dowry deaths, domestic violence, trafficking, and assault in a detailed manner per district and state. The report is a vital source of information for the assessment of safety, the efficiency of the administration, the state of public welfare, and the provision

of gender justice. Manually working through such vast and spanning over years datasets is not easy; hence, data visualisation tools like Power BI are indispensable in their role of making this information comprehensible for the local decision-makers.

Through the data analysis of crime for the years starting from 2017, the project unveils changes in the number of reported cases, areas with the highest risk, and new trends.

2. The Analysis Need:

Despite different legal reforms, the number of crimes against women is still increasing in various places. Policymakers very often do not have the necessary insights on a district level for interventions that are specifically aimed at. This analysis by:

- **Illustrating the growth patterns year-wise through different categories.**
- **Bringing to the forefront states and districts with consistently high incidents consistently.**
- **Supporting resource allocation with visual evidence (e.g., police, counselling centres).**
- **Increasing transparency and facilitating data-driven decision-making.**

Through interactive charts, KPIs, and drill-down maps, Power BI visualisations and DAX measures make it easier for the stakeholders to interpret the data.

3. Motivation Behind Choosing the Report / Area:

The issue of crimes against women brings to the fore the question of safety, the right of women to economically participate, and even the very national development. Since this is a social issue, it would be appropriate to say that it is very much in line with the UN Sustainable Development Goals (SDGs), particularly SDG 5 (Gender Equality), SDG 11 (Sustainable Cities & Communities), and SDG 16 (Peace, Justice & Strong Institutions).

The topic has been chosen for its social magnitude, public attention, and impacts on governance in India. The structured government dataset that was available for this study further allowed for a rigorous and non-partisan approach to the analysis.

Dataset Source: District-wise Crime Against Women (2017 onwards) – Government Open Data Repository.

4. Objective of the Analysis:

The main objectives of the project are:

- **To make a year-wise analysis of the growth and crime patterns against women in India.**
- **To examine different regions' performance and to point out the areas with consistently high or low rates.**
- **To obtain significant understandings utilising Power BI dashboards and DAX-determined metrics.**
- **To suggest policy-oriented measures based on the trends seen in the data.**

5. Report Details:

- **Report Name: Dataset of Crime Against Women by District and State (2017 Onwards)**
- **Source: Open Data Portal of the Government**
- **File Used: districtwise-crime-against-women-2017-onwards.csv**
- **Sections Extracted:**
    - **Annual crime sums**
    - **Crime types (Rape, Dowry, Kidnapping, Assault, etc.)**
    - **Mapping of district names to states**
    - **Total crime counts, along with metrics derived**

6. Data Processing Steps:

a. Data Extraction

- **CSV dataset imported from the Open Government Data (OGD) platform.**
- **Columns extracted: Year, State, District, Crime Category, Crime Count.**

b. Data Cleaning

- **Removed null and duplicate values.**
- **Standardised state/district names.**
- **Converted whole numbers to decimal numbers.**
- **Handled missing years by filling zeros where required.**

c. Data Modelling

- **Relationships established using surrogate keys.**
- **DAX Measures used for:**
  - **Total Crimes**

```
Total Crimes =
    SUM ( CrimeData[murder_with_rape_gang_rape] ) +
    SUM ( CrimeData[dowry_deaths] ) +
    SUM ( CrimeData[abetment_to_suicide_of_women] ) +
    SUM ( CrimeData[miscarriage] ) +
    SUM ( CrimeData[acid_attack] ) +
    SUM ( CrimeData[attempt_to_acid_attack] ) +
    SUM ( CrimeData[cruelty_by_husband_or_his_relatives] ) +
    SUM ( CrimeData[kidnapping_and_abduction] ) +
    SUM ( CrimeData[kidnapping_abduction_in_order_to_murder] ) +
    SUM ( CrimeData[kidnapping_for_ransom] ) +
    SUM ( CrimeData[kidnp_and_abductn_of_women_above_18_for_marrg] ) +
    SUM ( CrimeData[kidnp_and_abductn_of_girls_below_18_for_marrg] ) +
    SUM ( CrimeData[procuration_of_minor_girls] ) +
    SUM ( CrimeData[importation_of_girls_from_foreign] ) +
    SUM ( CrimeData[kidnapping_and_abduction_of_women_others] ) +
    SUM ( CrimeData[human_trafficking] ) +
    SUM ( CrimeData[selling_of_minor_girls] ) +
    SUM ( CrimeData[buying_of_minor_girls] ) +
    SUM ( CrimeData[rape_women_above_18] ) +
    SUM ( CrimeData[rape_girls_below_18] ) +
    SUM ( CrimeData[attempt_to_commit_rape_above_18] ) +
    SUM ( CrimeData[attempt_to_commit_rape_girls_below_18] ) +
    SUM ( CrimeData[assault_on_womenabove_18] ) +
    SUM ( CrimeData[assault_on_women_below_18] ) +
    SUM ( CrimeData[insult_to_the_modesty_of_women_above_18] ) +
    SUM ( CrimeData[insult_to_the_modesty_of_women_below_18] ) +
    SUM ( CrimeData[dowry_prohibition] ) +
    SUM ( CrimeData[procuring_inducing_children_for_the_sake_of_prostitution] ) +
    SUM ( CrimeData[detaining_a_prsn_in_premises_where_prost_is_carried] ) +
    SUM ( CrimeData[prostitution_in_or_in_the_vicinity_of_public_places] ) +
    SUM ( CrimeData[seducing_or_soliciting_for_purpose_of_prostitution] ) +
    SUM ( CrimeData[other_sections_under_itp_act] ) +
```

- **Total Cyber Crimes**

```
Total Cyber Crimes =
    SUM ( CrimeData[publshng_or_transmitting_of_sexually_explicit_mtrl] ) +
    SUM ( CrimeData[other_women_centric_cyber_crimes] ) +
    SUM ( CrimeData[use_of_child_for_pornography] )
```

- **Total Domestic Cruelty**

```
Total Domestic Cruelty =
    SUM ( CrimeData[cruelty_by_husband_or_his_relatives] ) +
    SUM ( CrimeData[protection_of_women_from_domestic_violence] )
```

- **Total Dowry Related**

```
Total Dowry Related =
    SUM ( CrimeData[dowry_deaths] ) +
    SUM ( CrimeData[dowry_prohibition] )
```

- **Total Kidnapping & Abduction**

```
Total Kidnapping & Abduction =
    SUM ( CrimeData[kidnapping_and_abduction] ) +
    SUM ( CrimeData[kidnapping_abduction_in_order_to_murder] ) +
    SUM ( CrimeData[kidnapping_for_ransom] ) +
    SUM ( CrimeData[kidnp_and_abductn_of_women_above_18_for_marrg] ) +
    SUM ( CrimeData[kidnp_and_abductn_of_girls_below_18_for_marrg] ) +
    SUM ( CrimeData[kidnapping_and_abduction_of_women_others] )
```

- **Total POSCO**

```
Total POCSO =
    SUM ( CrimeData[child_rape] ) +
    SUM ( CrimeData[sexual_assault_of_children] ) +
    SUM ( CrimeData[child_sexual_harassment] ) +
    SUM ( CrimeData[offences_of_pocso_act] ) +
    SUM ( CrimeData[pocso_act_unnatural_offences] )
```

- **Total Rape**

```
Total Rape =
    SUM ( CrimeData[rape_women_above_18] ) +
    SUM ( CrimeData[rape_girls_below_18] ) +
    SUM ( CrimeData[child_rape] )
```

- **Category Rank**

```
Category Rank =
RANKX(
    ALL(CrimeCategory[Category]),
    [Crime Category Value],
    BLANK(),
    DESC,
    Skip
)
```

- **Crime Category Value**

```
Crime Category Value =
SWITCH(
    SELECTEDVALUE(CrimeCategory[Category]),
    "Total Rape", [Total Rape],
    "Total Kidnapping & Abduction", [Total Kidnapping & Abduction],
    "Total Domestic Cruelty", [Total Domestic Cruelty],
    "Total Dowry Related", [Total Dowry Related],
    "Total Cyber Crimes", [Total Cyber Crimes],
    "Total POCSO", [Total POCSO]
```

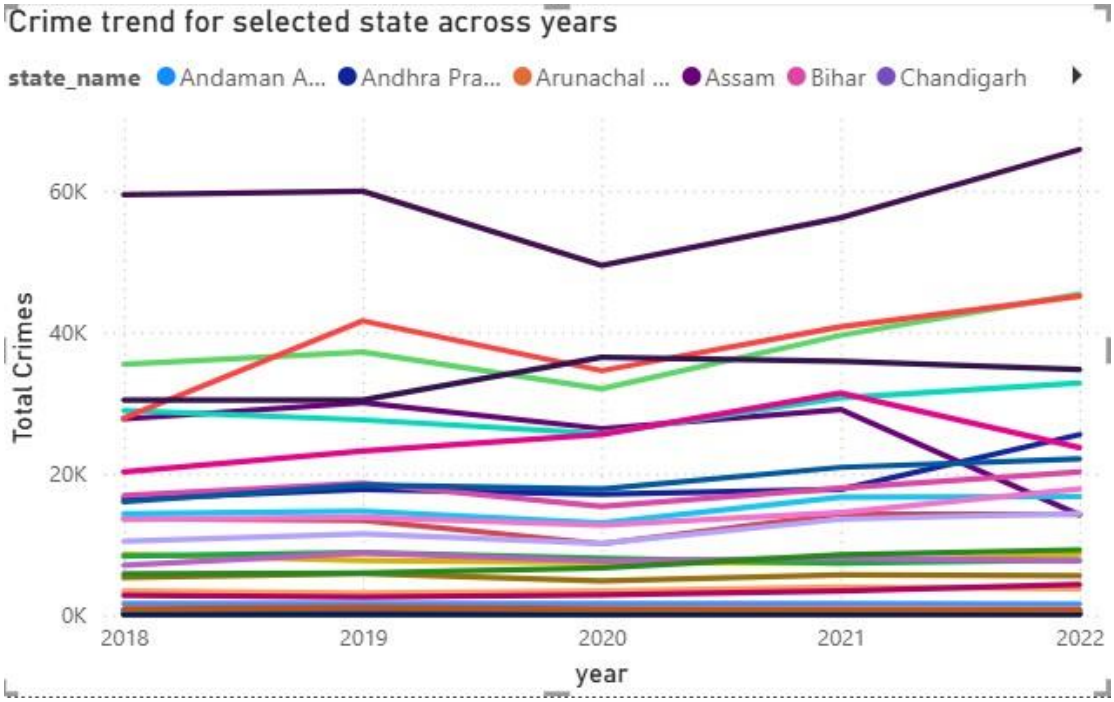
7. Analytical Graphs & Interpretation:

Power BI Visuals:

1. **KPI Panel – Total Crimes, YoY %, and the State with the Most Crimes**

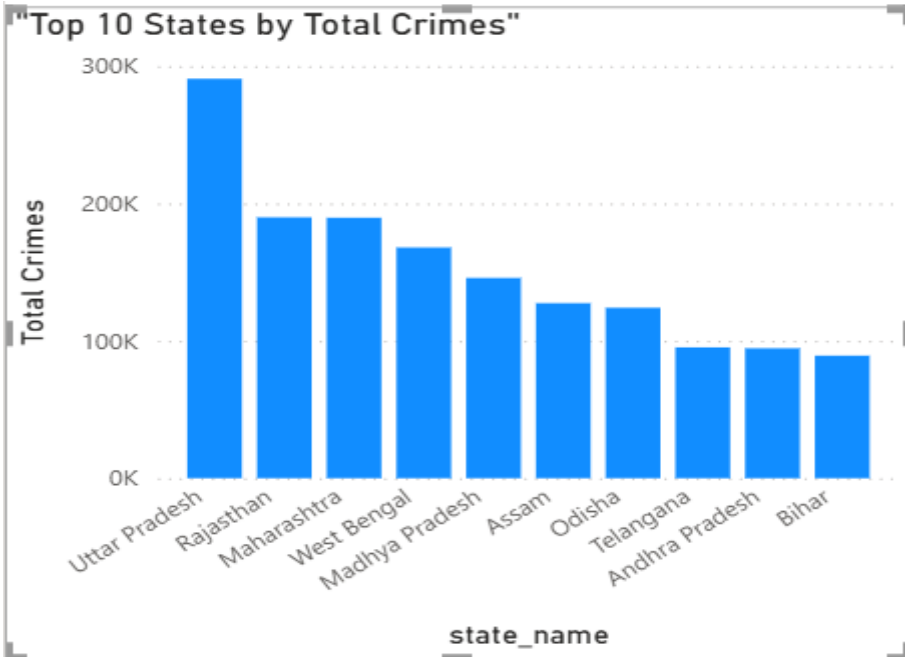
- **Insight: There are some states that report an increase in the number of incidents, as they are now being reported or the situation is getting worse.**

2. **Yearly Trend Line**



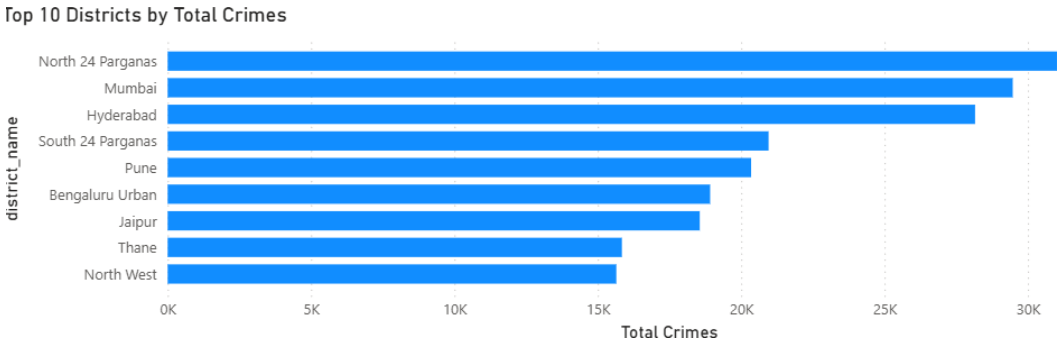
- **Insight: There is a noticeable increase in many categories after 2019.**

3. **State-wise Bar Graph Comparison**



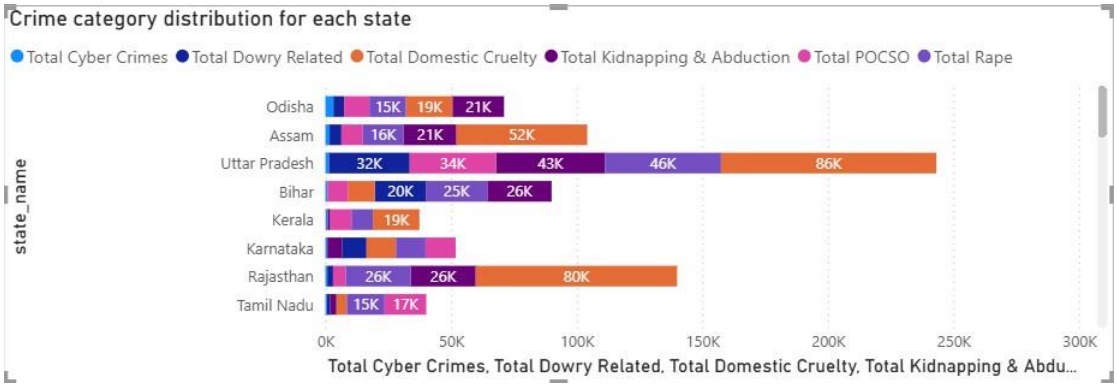
- **Insight: UP, Rajasthan, and Maharashtra are the states that most often report a large number of cases.**

4. District-wise Ranking Chart



- **Insight: Certain districts are always among the top 10 places with cases, thus indicating a long-term risk.**

5. Category-wise Stacked Chart



- **Insight: The domestic violence and attacks on women are to blame for most of the cases in the different areas.**

The above-mentioned visual aids are instrumental in unravelling the patterns of crime, determining the hotspots and tracing the movement of cases throughout India.

8. Further Scope of Work:

- **Incorporating historical NCRB data for extensive trend analysis.**
- **Inclusion of predictive models (ARIMA, Prophet) for estimating the future crime rate.**
- **Application of text mining on the FIR summaries through NLP to derive qualitative insights.**
- **Merging of socio-economic indicators (literacy, income, police strength) for correlation analysis.**
- **Establishment of automated alerts or an AI-based risk scoring system.**

9. Conclusion:

Power BI effectively transformed public crime datasets into actionable insights, and this project demonstrates it. The study gives out the main patterns of crimes against women, identifies the areas that are at risk the most, and shows the differences between regions in India. The use of the dashboard to depict complicated datasets straightforwardly makes it easier for decision-makers, researchers, and social workers to grasp the patterns and come up with treatments that are properly focused. The project improved the capabilities related to data extraction, cleansing, modelling, DAX calculations, and visualisation. In short, this analysis confirms that the power of data-driven decision-making could result in a positive social change.

10. Annexure:

- **Snapshot of raw CSV data**
- **Power BI visuals snapshots**
- **DAX formulas used**